Foundations of Autoregressive model

[Autoregressive model]

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Using the recursion formula yields ρ 2 = γ 2 / γ 0 = ϕ 1 2 − ϕ 2 2 + ϕ 2 1 − ϕ 2 {\displaystyle \rho _{2}=\gamma _{2}/\gamma _{0}={\frac {\varphi _{1}^{2}-\varphi _{2}^{2}+\varphi _{2}}{1-\varphi _{2}}}}

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Implementations in statistics packages R – the stats package includes ar function; the astsa package includes sarima function to fit various models including AR. MATLAB – the Econometrics Toolbox and System Identification Toolbox include AR models. MATLAB and Octave – the TSA toolbox contains several estimation functions for uni-variate, multivariate, and adaptive AR models. PyMC3 – the Bayesian statistics and probabilistic programming framework supports AR modes with p lags. bayesloop – supports parameter inference and model selection for the AR-1 process with time-varying parameters. Python – statsmodels.org hosts an AR model.

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Φ (ω) = 1 2 π ∑ n = − ∞ ∞ B n e − i ω n = 1 2 π (σ ε 2 1 + ϕ 2 − 2 ϕ cos ■ (ω)) . {\displaystyle \Phi (\omega)={\frac {1}{\sqrt {2\pi }}}\,\sum _{n=-\infty }^{\infty }B_{n}e^{-i\omega n}={\frac {1}{\sqrt {2\pi }}}\,\left({\frac {\sigma _{\varepsilon }^{2}}{1+\varphi ^{2}-2\varphi \cos(\omega)}}\right).}

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which can be solved for all { ϕ m ; m = 1 , 2 , … , p } . {\displaystyle \{\varphi _{m};m=1,2,\dots ,p\}.} The remaining equation for m = 0 is

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The simplest AR process is AR(0), which has no dependence between the terms. Only the error/innovation/noise term contributes to the output of the process, so in the figure, AR(0) corresponds to white noise. For an AR(1) process with a positive ϕ {\displaystyle \varphi } , only the previous term in the process and the noise term contribute to the output. If ϕ {\displaystyle \varphi } is close to 0, then the process still looks like white noise, but as ϕ {\displaystyle \varphi } approaches 1, the output gets a larger contribution from the previous term relative to the noise. This results in a "smoothing" or integration of the output, similar to a low pass filter. For an AR(2) process, the previous two terms and the noise term contribute to the output. If both ϕ 1 {\displaystyle \varphi _{1}} and ϕ 2 {\displaystyle \varphi _{2}} are positive, the output will resemble a low pass filter, with the high frequency part of the noise decreased. If ϕ 1 {\displaystyle \varphi _{1}} is positive while ϕ 2 {\displaystyle \varphi _{2}} is negative, then the process favors changes in sign between terms of the process. The output oscillates. This can be linked to edge detection or detection of change in direction.

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An autoregressive model can thus be viewed as the output of an all-pole infinite impulse response filter whose input is white noise. Some parameter constraints are necessary for the model to remain weak-sense stationary. For example, processes in the AR(1) model with | ϕ 1 | ≥ 1 {\displaystyle |\varphi _{1}|\geq 1} are not stationary. More generally, for an AR(p) model to be weak-sense stationary, the roots of the polynomial Φ (z) := 1 − ∑ i = 1 p ϕ i z i {\displaystyle \Phi (z):=\textstyle 1-\sum _{i=1}^{p}\varphi _{i}z^{i}} must lie outside the unit circle, i.e., each (complex) root z i {\displaystyle z_{i}} must satisfy | z i | > 1 {\displaystyle |z_{i}|>1} (see pages 89,92).

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In statistics, an autoregressive (AR) model is a modelled representation of a type of random process. It can be used to describe time-varying processes from many natural and artificial sources. The model specifies output variables that are dependent linearly on their own previous values on a stochastic basis. The model is in the form of a stochastic difference equation (or recurrence relation) which should not be confused with a differential equation. Together with the moving-average (MA) model, it is a special case and key component of the more general autoregressive–moving-average (ARMA) and autoregressive integrated moving average (ARIMA) models of time series, which have a more complicated stochastic structure; it is also a special case of the vector autoregressive model (VAR), which consists of a system of more than one interlocking stochastic difference equation in more than one evolving random variable. Another important extension is the time-varying autoregressive (TVAR) model, where the autoregressive coefficients are allowed to change over time to model evolving or non-stationary processes. TVAR models are widely applied in cases where the underlying dynamics of the system are not constant, such as in sensors time series modelling, climate science, economics and finance (as econometrics), signal processing, telecommunications, radar systems, and biological signals. Unlike the moving-average (MA) model, the autoregressive model is not always stationary; non-stationarity can arise either due to the presence of a unit root or due to time-varying model parameters, as in time-varying autoregressive models. Large language models are called autoregressive, but they are not a classical autoregressive model in this sense because they are not linear.

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where ϕ 1 , … , ϕ p {\displaystyle \varphi _{1},\ldots ,\varphi _{p}} are the parameters of the model, and ε t {\displaystyle \varepsilon _{t}} is white noise. This can be equivalently written using the backshift operator B as

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This expression is periodic due to the discrete nature of the X j {\displaystyle X_{j}} , which is manifested as the cosine term in the denominator. If we assume that the sampling time (Δ t = 1 {\displaystyle \Delta t=1}) is much smaller than the decay time (τ {\displaystyle \tau }), then we can use a continuum approximation to B n {\displaystyle B_{n}} :

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γ m = ∑ k = 1 p ϕ k γ m − k + σ ε 2 δ m , 0 , {\displaystyle \gamma _{m}=\sum _{k=1}^{p}\varphi _{k}\gamma _{m-k}+\sigma _{\varepsilon }^{2}\delta _{m,0},}

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The terms involving square roots are all real in the case of complex poles since they exist only when ϕ 2 < 0 {\displaystyle \varphi _{2}<0} . Otherwise the process has real roots, and:

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Impulse response The impulse response of a system is the change in an evolving variable in response to a change in the value of a shock term k periods earlier, as a function of k. Since the AR model is a special case of the vector autoregressive model, the computation of the impulse response in vector autoregression#impulse response applies here.